

Natural Language Processing

Trainer: Prashant Nair

What is NLP?

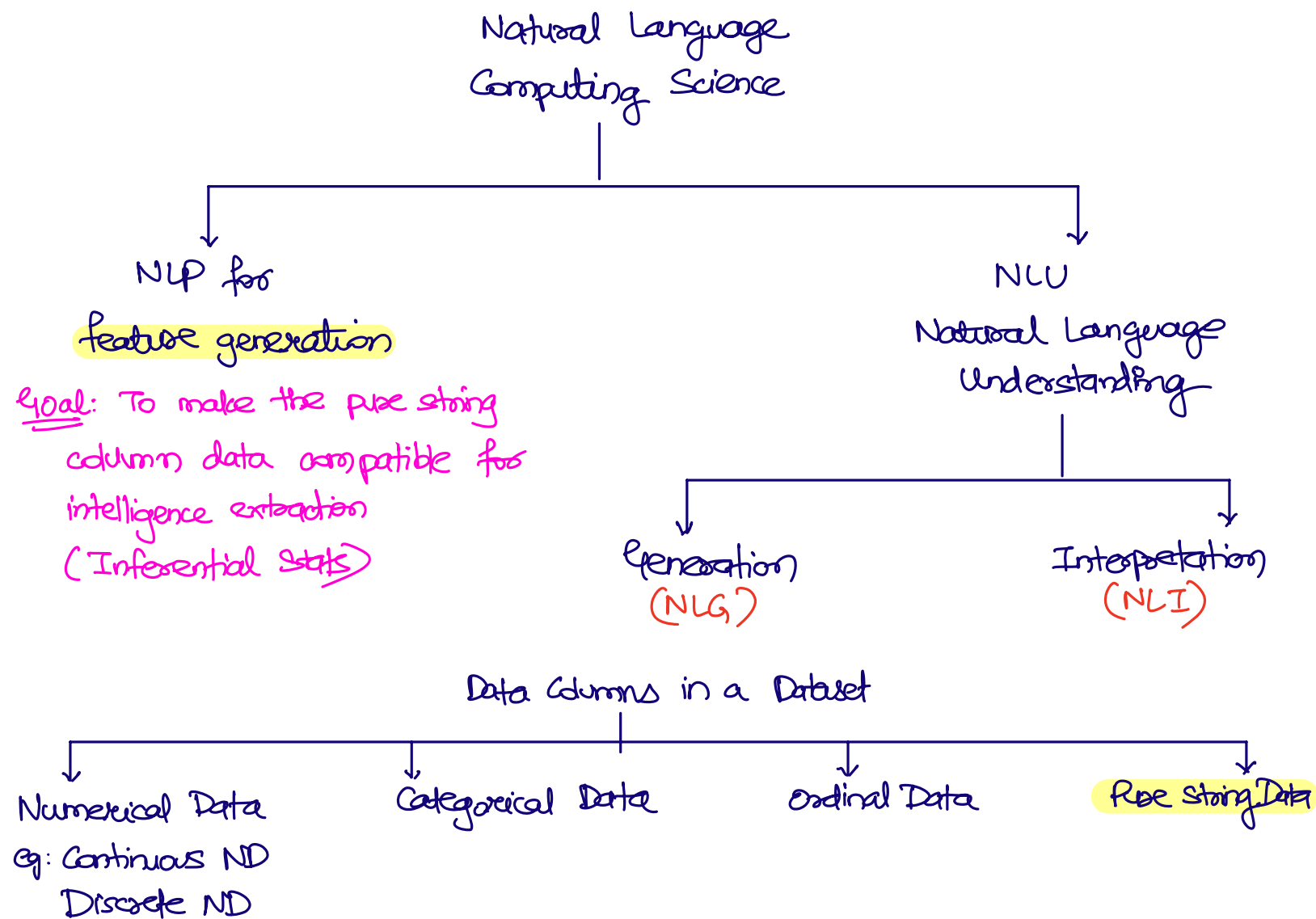
NLP is the branch of AI that helps

UNDERSTAND

INTERPRET

MANIPULATE

the human language.



Text Preprocessing Stage in NLP for feature generation

It has 8 stages. The stages can be skipped if not required for the problem statement.

① Data Acquisition [Capture raw text data]

- file
- database
- online [web services: XML, JSON, HTML]
- Scraping.

This stage is all about getting raw data and storing the same in a persistent storage.

② Cleaning stage:

This stage is all about extracting required data from raw data.

Eg: file.html

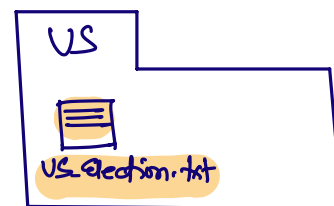
```
<html>
<head>
  <title> US Election </title>
</head>
<body>
  <p> ===== </p>
</body>
</html>
```

Option 1:
Define structure
and separate data
from metadata.

→

country	title	article
US	USElection	=====

Option 2:



② **Normalization**: Bringing the feature col data in one scale.

[Text Normalization]

The goal of text normalization is to **unify** the text data.

↑ converting text data in one single format

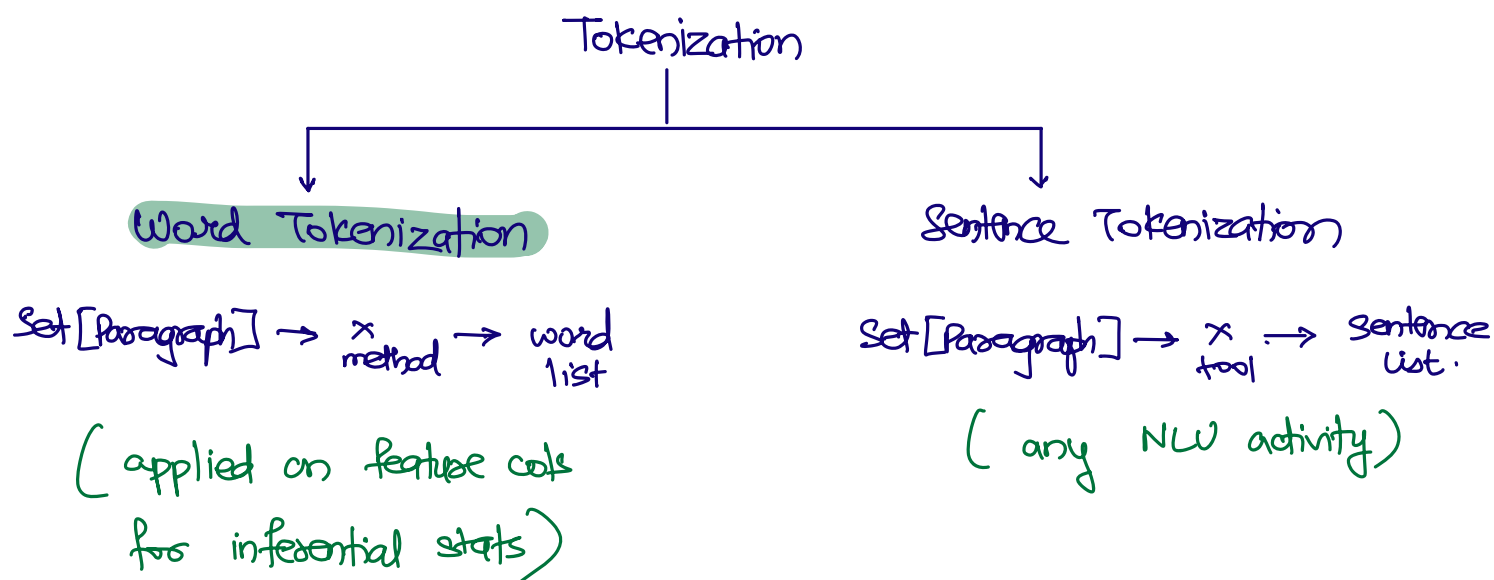
eg: ① making text follow one case

[lowercase]*

[uppercase]

② Removal of Punctuations.*

④ Tokenization: Convert text data into a sequence of tokens.



⑤ **Stopwords** Removal:

↑ those words that has least/no statistical impact on the AI algo.

(outliers in text processing)

eg: Bruno is my best dog!

↓ Normalization

bruno is my best dog

↓ stopwords

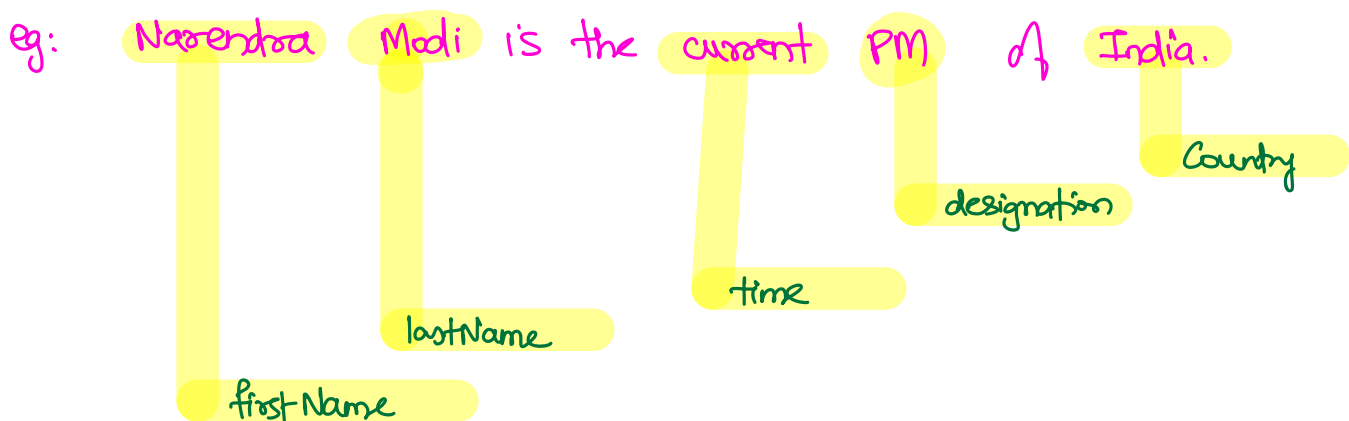
bruno best dog.

⑥ POS tagging [Parts of Speech].

The goal of POS tagging is to identify how the words are used in a sentence such that the system can understand the CONTEXT of the sentence.

⑦ NER [Named Entity Recognition].

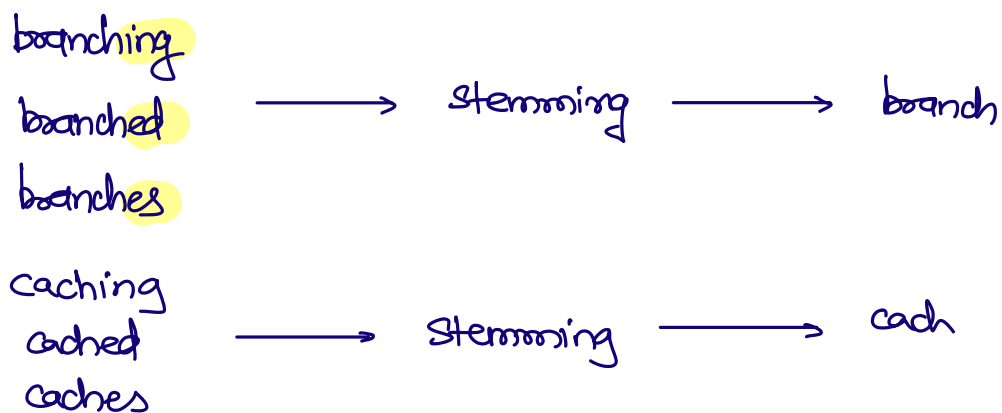
The goal of NER is to recognize the entity of the word.



⑧ Stemming and Lemmatization

Advanced form of Normalization

The goal of stemming and lemmatization is to identify the ROOT word.



catching
cached
caches

→ Lemma → cache

Lemmatization is all about identifying the root word ensuring the root word is the part of the language dictionary.